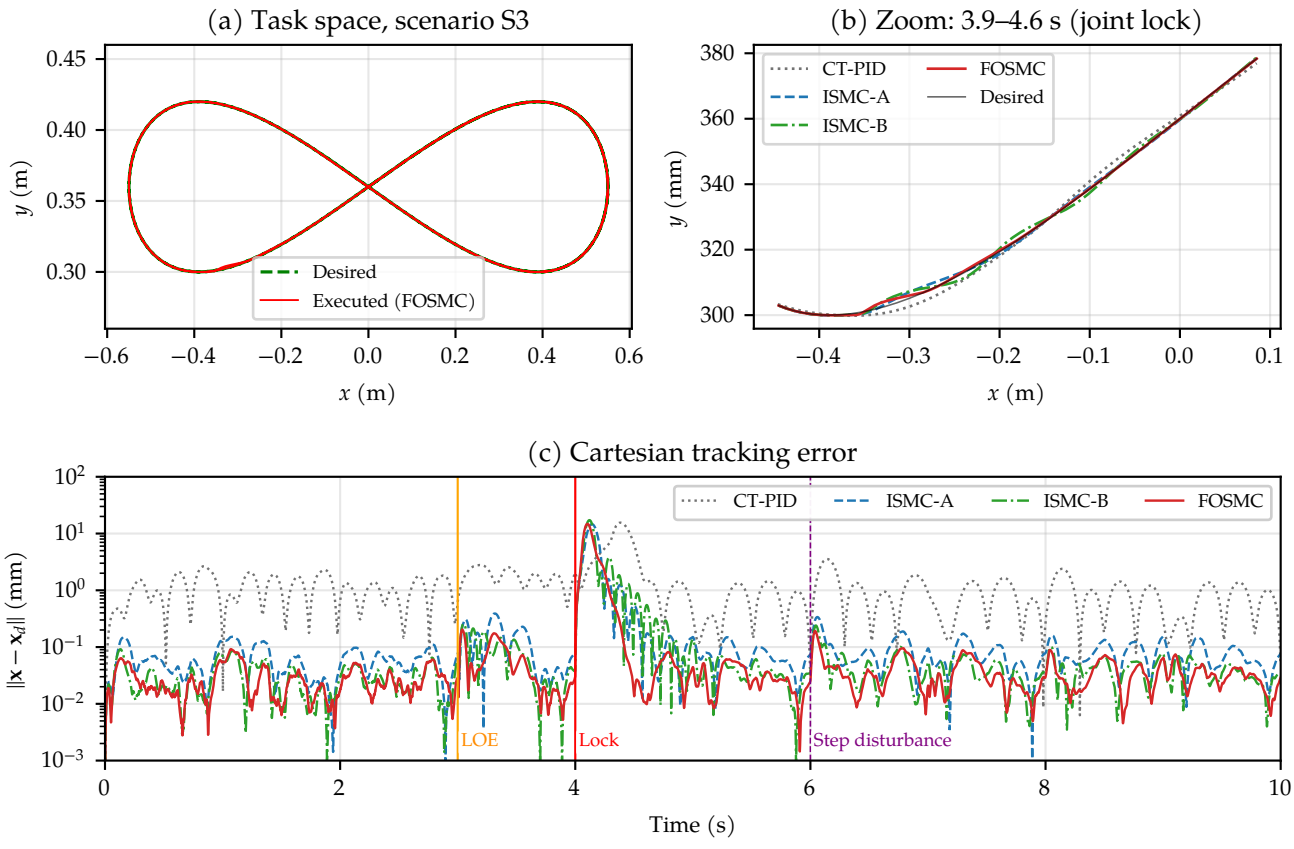

(a) Task space, scenario S3
0.45
0.40
0.35
0.30
y (m)
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
x (m)
Desired
Executed (FOSMC)
(b) Zoom: 3.9–4.6 s (joint lock)
380
360
340
320
300
y (mm)
−0.4
−0.3
−0.2
−0.1
0.0
0.1
x (m)
CT-PID
ISMC-A
ISMC-B
FOSMC
Desired
(c) Cartesian tracking error
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
$\|\mathbf{x}-\mathbf{x}_d\|$ (mm)
CT-PID
ISMC-A
ISMC-B
FOSMC
LOE
Lock
Step disturbance
0
2
4
6
8
10
Time (s)